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Προπτυχιακό Πρόγραμμα Σπουδών
2024-5

Ασφάλεια Δικτύων
Εργαστηριακή Άσκηση 2
Ατομική – στον υπολογιστή σας

Instructions for ARP Poisoning Exercise (AUEB)

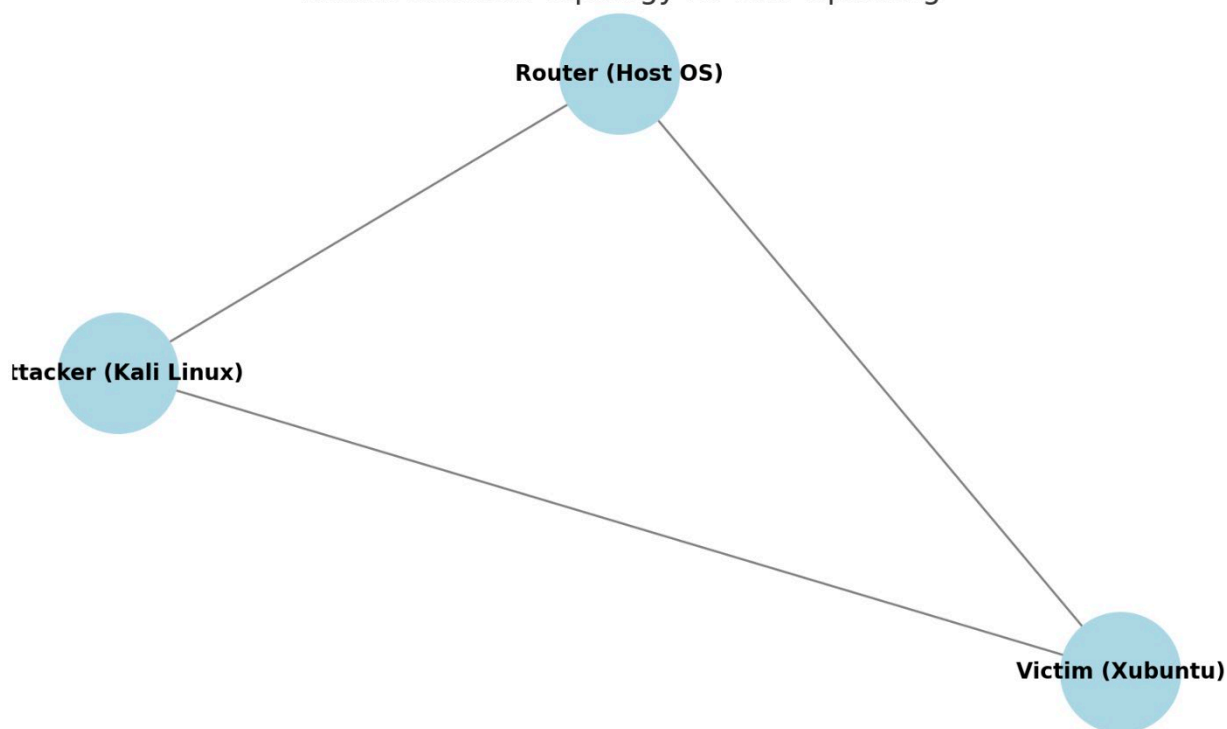
Prerequisites

1. **Virtualization Software:** Install VirtualBox or VMware.
2. **Operating Systems:**
 - **Attacker:** Any new version of Kali Linux
 - **Victim:** Any new version of Ubuntu
3. **Network Tools:**
 - **Attacker Machine:** Pre-installed ARP spoofing tools like **arp spoof** from the Dsniff suite, **urlsnarf**, and Wireshark.

Step 1: Set Up Virtual Network

1. **Create a Custom NAT Network:**
 - Open VirtualBox, go to **File > Preferences > Network**.
 - Click on the **NAT Networks** tab and create a new NAT network.
 - Assign an IP range (e.g., **192.168.1.0/24**).
 - Ensure **DHCP Server** is disabled for manual IP assignment.
2. **Assign Virtual Machines to the NAT Network:**
 - Go to **Settings > Network** for each VM.
 - Set the adapter type to **NAT Network** and select the custom NAT network.

Virtual Network Topology for ARP Spoofing



Step 2: Configure IP and MAC Addresses

Assign static IPs to each machine. Example configuration:

Machine	IP Address	MAC Address
Attacker (Kali)	192.168.1.10	08:00:27:44:F9:12
Victim (Ubuntu)	192.168.1.11	08:00:27:57:29:69
Router (Host OS/Virtual Router)	192.168.1.1	Auto-assigned

Step 3: Configure the Attacker Machine

Enable IP Forwarding: Run the following command to enable packet forwarding:

```
sudo sysctl -w net.ipv4.ip_forward=1
```

Verify Network Tools: Ensure `arpspoof` and `urlsnarf` are installed:

```
sudo apt-get install dsniff
```

Objective of the Lab Exercise

- 1) Conduct a Man-in-the-Middle (MitM) attack using ARP Cache Poisoning (APR Spoof) in a Virtualized Environment.
- 2) Validate the attack is taking place correctly with relevant tools and services of your choice.

During the above objectives, take screenshots of the whole implementation for both the virtual lab setup and the MitM attack that capture utilized tools' commands, ARP tables (Before & After), and the validation process

Report Instructions

Prepare a comprehensive, structured report that documents the exercise's objectives and key findings.

1. **Document Length: 7 - 15 pages (Font 12)**
2. **Comprehensive Documentation**
 - Systematically describe each procedural step
 - Ensure thorough and precise explanation of methodologies employed
3. **Visual Documentation**
 - Incorporate high-quality screenshots
 - Provide detailed textual descriptions for each screenshot
 - Ensure screenshots are relevant, clear, and directly illustrative of key processes
 - Cite appropriate references where necessary
 - Maintain a structured and narrative flow

Deadline 17/1/25, 9pm